MOHAMMED MINHAJ MAHMOOD

Data Science & Analytics Engineer · ML | NLP | MLOps | A/B Testing

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PROFESSIONAL SUMMARY

Results-driven Data Science & Analytics Engineer with a B.Tech in Computer Science (Data Science) and a hands-on portfolio of 5 end-to-end ML projects covering customer segmentation, time series forecasting, NLP, causal inference, and MLOps. Proficient in the full data science lifecycle — from EDA and feature engineering to model deployment and monitoring. Adept at translating complex model outputs into measurable business impact. Seeking an Associate Data Scientist / Data Analyst role where I can apply statistical modelling, machine learning, and storytelling skills to solve real business problems.

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, SciPy), SQL (PostgreSQL, MySQL), R
Machine Learning: XGBoost, LightGBM, Random Forest, Logistic Regression, K-Means, SMOTE, SHAP
Deep Learning / NLP: PyTorch, HuggingFace Transformers, DistilBERT, BERTopic, SentenceTransformers, spaCy
Time Series: ARIMA, SARIMA, Prophet, LightGBM Forecasting, TimeSeriesSplit, pmdarima
Causal Inference: A/B Testing, CausalML, EconML, T-Learner, X-Learner, Qini Curves
MLOps & Deployment: MLflow, FastAPI, Docker, Docker Compose, GitHub Actions CI/CD, EvidentlyAI, Streamlit
Data Visualisation: Plotly, Matplotlib, Seaborn, Tableau, Power BI
Cloud & Tools: AWS (S3, EC2), Git, GitHub, Jupyter, VS Code

PROJECTS

Customer Segmentation & Churn Prediction — GitHub ↗

Tech: *Python · XGBoost · SMOTE · SHAP · K-Means · MLflow · Streamlit*

- Engineered a two-part customer intelligence system on 7,043 Telco records — RFM segmentation using K-Means (5 clusters, Silhouette score 0.61) and churn prediction using XGBoost (AUC 0.89, F1 0.82).
- Handled severe class imbalance (26.5% churn rate) using SMOTE, improving recall on the minority class from 61% to 81% without sacrificing precision.
- Applied SHAP explainability to identify top 5 churn drivers — contract type, tenure, and monthly charges — delivering actionable retention strategy directly to business stakeholders.
- Deployed interactive Streamlit dashboard enabling non-technical marketing teams to explore customer segments and predict individual churn risk in real time.
- Estimated business impact: 3.8x campaign ROI improvement and ₹14–18 Lakhs/month revenue saved per 10,000 customers through targeted retention over blanket campaigns.

Demand Forecasting & Time Series Analysis — GitHub ↗

Tech: *Python · ARIMA · SARIMA · Prophet · LightGBM · MLflow · FastAPI · Streamlit*

- Built a multi-model forecasting pipeline on UCI Bike Sharing dataset — benchmarked ARIMA, SARIMA, Prophet, and LightGBM using TimeSeriesSplit (5-fold) to prevent data leakage.
- Engineered 20+ lag, rolling statistics, and calendar features (day of week, holiday flags, temperature bins) for LightGBM, achieving MAPE of 6.8% vs ARIMA baseline of 14.2% — a 52% improvement.
- Tracked 40+ MLflow experiments across hyperparameter combinations; registered best model to MLflow Model Registry for versioned production deployment.
- Served real-time forecasts via FastAPI endpoint (sub-100ms response) and built Streamlit dashboard for business planning teams to explore demand scenarios interactively.
- Identified key demand drivers: weekly seasonality (weekday 2.3x weekend), 17% YoY growth trend, and 31% demand drop on public holidays — directly informing procurement decisions.

NLP Review Intelligence System — GitHub ↗

Tech: *Python · DistilBERT · HuggingFace Transformers · BERTopic · SentenceTransformers · spaCy · PyTorch*

- Fine-tuned DistilBERT on 50,000 Amazon reviews for 3-class sentiment classification, achieving 91.4% accuracy and 0.89 macro F1 — outperforming TF-IDF + Logistic Regression baseline by 13.1%.
- Applied BERTopic (UMAP + HDBSCAN + semantic embeddings) on 500,000 reviews to automatically discover 10 coherent product intelligence topics without any labelling.

- Identified battery life and delivery & packaging as the top two negative topics, with durability issues flagging 9,800 reviews — providing product and QA teams a direct action roadmap.
- Delivered 100% review coverage vs prior 1% manual sampling — effectively a 100x expansion in customer intelligence from the same data with zero additional manual effort.

A/B Testing & Uplift Modeling System — GitHub ↗

Tech: *Python · CausalML · EconML · SciPy · Statsmodels · Plotly · Streamlit*

- Conducted rigorous A/B test analysis on Hillstrom email campaign dataset using chi-square, t-tests, confidence intervals, and power analysis — distinguishing statistically significant results from noise.
- Built T-Learner, X-Learner, and Uplift Random Forest models using CausalML; X-Learner achieved AUUC of 0.739 — 47.8% above random baseline targeting.
- Ranked all customers by uplift score into deciles; top 20% of customers generated 52% of total campaign value, while bottom 20% showed -19% uplift (Do Not Disturb segment).
- Demonstrated that targeting the top 38% of customers by uplift score delivered 85.5% of campaign revenue at 38% of the cost — a 41% improvement in campaign ROI (1.48x → 2.08x).

End-to-End ML Pipeline (MLOps) — GitHub ↗

Tech: *Python · FastAPI · Docker · GitHub Actions · MLflow · EvidentlyAI · Pytest · Render*

- Architected a production-grade MLOps pipeline: FastAPI REST API (38ms avg response, /predict, /predict/batch, /health endpoints) with Pydantic v2 input validation and auto-generated OpenAPI docs.
- Containerised API and Streamlit dashboard as separate Docker services orchestrated via docker-compose, eliminating all environment inconsistencies across development and production.
- Built GitHub Actions CI/CD pipeline: automated pytest → Docker build → Docker Hub push → auto-deploy to Render on every push to main — reducing deployment time from 3 days to 4 minutes.
- Implemented EvidentlyAI weekly drift monitoring with automated HTML reports comparing training vs production data distributions — enabling proactive model retraining before accuracy degrades.
- Achieved 87% test coverage, 99.7% API uptime, and full MLflow experiment tracking across all model versions — establishing a reusable MLOps template applicable to any future project.

WORK EXPERIENCE

Business Development Associate | Syncpedia Technologies

Jan 2026 – Jun 2026

EdTech startup · B2B lead generation · Student outreach campaigns

- Led end-to-end B2B lead generation campaign targeting Computer Science students at Vaagdevi College of Engineering, Warangal — managing cold outreach, qualification, and pipeline tracking.
- Designed and executed structured cold calling framework, achieving consistent daily outreach targets and maintaining detailed CRM records for follow-up tracking.
- Developed outreach messaging that improved student response rates; contributed to refining the company's student acquisition playbook based on real campaign learnings.

EDUCATION

B.Tech — Computer Science Engineering (Data Science)

2022 – 2026

Vaagdevi College of Engineering · Warangal, Telangana

- Relevant coursework: Machine Learning, Data Mining, Statistical Analysis, Database Systems, Python Programming, Operations Research

Intermediate Education (MPC)

2020 – 2022

SR Junior College · Warangal, Telangana

Secondary School Certificate

Graduated 2020

Suffah High School · Warangal, Telangana

CERTIFICATIONS & ADDITIONAL

- Data Science with Python — Codebasics / Atliq (In Progress, 2026)
- Machine Learning Specialization — Andrew Ng, Coursera (2025)

- AWS Cloud Practitioner Essentials (2025)
- Languages: English (Professional), Telugu (Native), Hindi (Conversational), Urdu (Native)
- Open to roles: Data Analyst, Associate Data Scientist, ML Engineer (Associate), Business Analyst
- Open to relocate anywhere in India